

Northwind Home: proving the complaints agent is worth extending

Domain FDE Pre-Work Deep Track • Blue Data Consulting • Stage 3: Measure

Where Day 2 left us

complaints_v2_decision is now safe: five rules, checked before every action, with anything risky escalated to a named person. The team lead wants it on all 400 complaints a week. Before anyone agrees, three questions have arrived that nobody can answer:

- **Finance:** what does one decision cost, and what will 400 a week cost?
- **Complaints team lead:** how many cases will the agent send back to my six people?
- **Head of Customer Experience:** are customers actually better off, or just handled faster?

Nothing records model cost or response time. Blocked attempts only appear in the terminal. And nobody has written down what “resolved” means, so two people counting the same week would get different answers.

Today’s decision

What the agent is trying to achieve, what it must never exceed, and what we only watch: written as numbers that any two people would calculate the same way.

Accuracy is not the goal. An agent can be right more often and still cost more, keep customers waiting, or push every hard case back to the team.

The five measures

Every number has one job: the agent **chases** it (Goal), must **respect** it (Limit), or we only **watch** it.

#	Metric	Job	Budget
1	Correct autonomous resolution: complaints the agent closed itself, with no block and no escalation	Goal	As high as possible, within the limits
2	Escalation rate: complaints sent to a person	Limit	No more than 30%
3	Must-escalate coverage: vulnerable or legal complaints that reached a person	Limit	100%, no exceptions
4	Cost per decision: model cost of handling one complaint	Limit	No more than £0.02
5	Response time: 95th percentile, per complaint	Limit	No more than 15 seconds

Watch only: steps per complaint • blocked attempts • tokens per decision.

The budgets are assumptions waiting for the team lead’s and Finance’s sign-off. The definitions are what you will argue about.

What to watch

The terminal shows the same three lines as Day 2. New today is the scorecard, printed by `metrics.py`. One BREACH means NOT READY:

Scorecard result	Meaning
GOAL	The number the agent is trying to improve: higher is better, and there is no pass mark
PASS	The metric is within its budget
BREACH	A limit was crossed: the agent is not ready to extend, however good the goal looks
WATCH	Shown for diagnosis only; it has no budget and nothing fails on it

What was built

complaints_v3_measured behaves exactly like Day 2: same rules, tools and instruction. What changed is what gets recorded and counted.

	complaints_v2_decision (Day 2)	complaints_v3_measured (Day 3)
Behaviour	Five rules, guardrail, automatic escalation	Identical
What is recorded	Escalations only, in escalations.log	Every model call, tool call, block and escalation, with tokens and time: recorder.py writes runs.jsonl
What success means	Not written down	metrics.py: five metrics, each with a contract (definition, grain, source, freshness, owner)
Budgets	None	Set in metrics.py and checked on every scorecard
Proof	5 decision scenarios	Also 5 hand-made cases in tests/sample_cases.csv, with each metric's expected value worked out by hand

Your task: 50 minutes, in pairs

CXM pairs. Work in usecase01_day3. Every command is in README-training-account.md. Record everything in Day3_Worksheets_UC01.docx. Do not change the budgets; propose changes in ADR-3.

#	Part	Time	What you do
1	Measure it cold	15 min	The opening questions, then your KPI tree split into Goal, Limit and Watch. Before opening the code.
2	Contract it	10 min	Write the contract for each metric. Work out by hand what each should be for the 5 sample cases.
3	Run it	15 min	README steps 8 to 16: the metric tests, the four Day 2 prompts on v3, your scorecard from runs.jsonl, then the prompts that move each metric.
4	Prove it	10 min	Map each metric to the data it needs, write ADR-3, then answer the seven questions below.

Seven questions

1. What is this agent's objective, in one sentence? Why is it not accuracy?
2. For each metric: does the agent chase it, respect it, or do we only watch it? What goes wrong if escalation rate became a Goal?
3. C-104 was escalated twice in one session. Does escalation rate count complaints or escalations? Who decides?
4. "Resolved" means closed with no block and no escalation. How would you know it was not reopened two weeks later?
5. Should cost per decision include the team lead's time on escalations? Which budget breaks first at 400 complaints a week?
6. Where did your hand-worked value differ from metrics.py? Was the code wrong, or the contract unclear?
7. Which metric needs data the agent does not have today, such as customers, reopened complaints or refunds? Where would it come from?

Bring to the debrief

The metric definition your pair disagreed on · the budget you would breach first · the metric you would move from Watch to Limit.

All data is made up. Nothing changes in any real system. If the environment breaks, hand up.